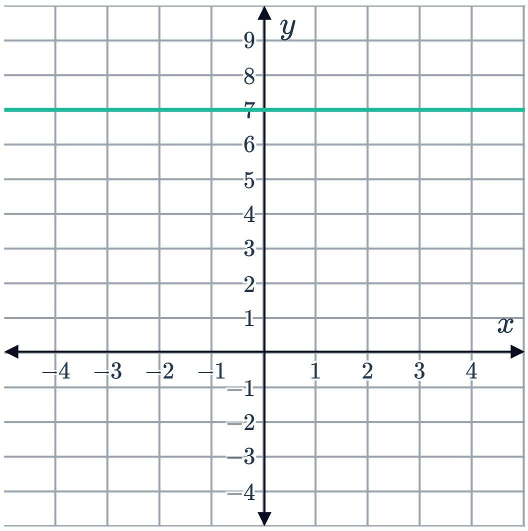
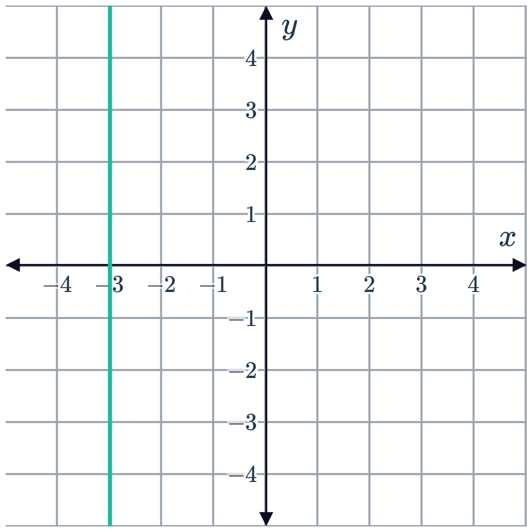
Gradient of a line

1 Describe the gradient of the lines in the following graphs:

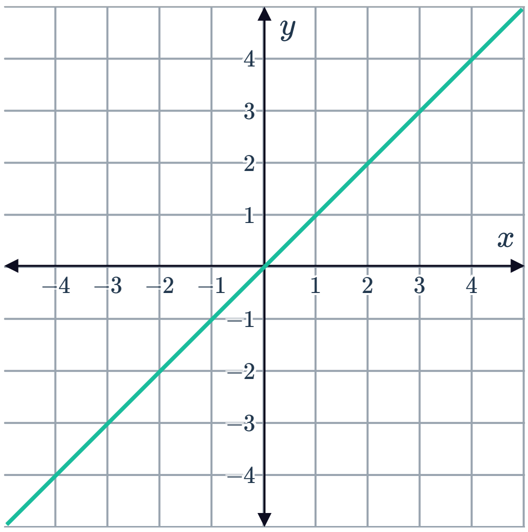
a



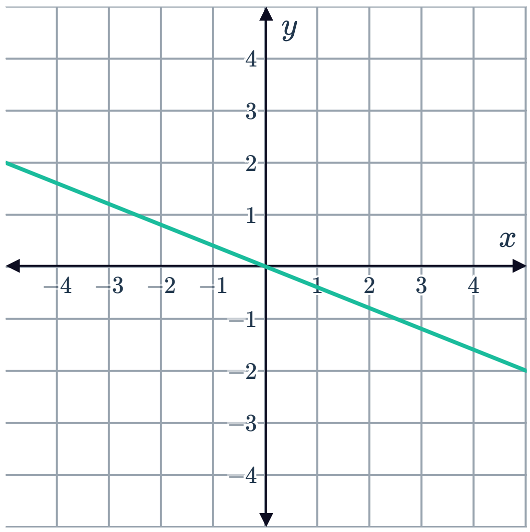
b



c



d



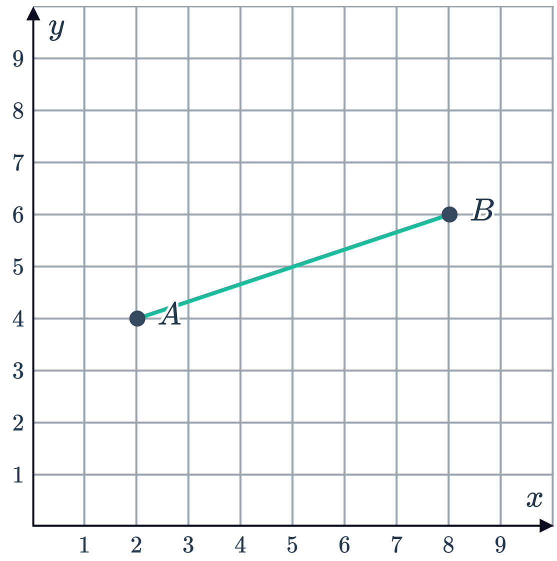
2 For each of the following intervals betwee[redacted]nts A and B:

i Find the rise.

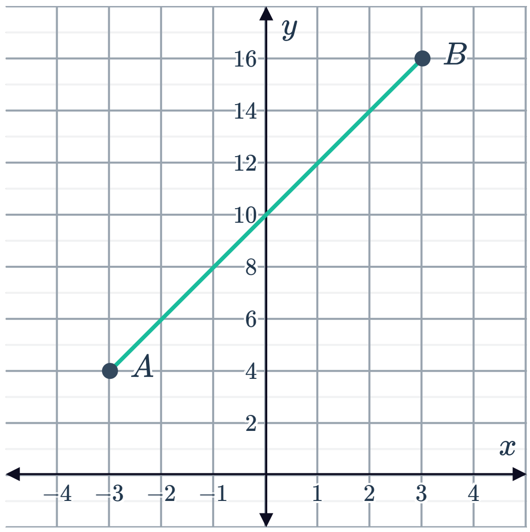
ii Find the run.

iii Find the gradient.

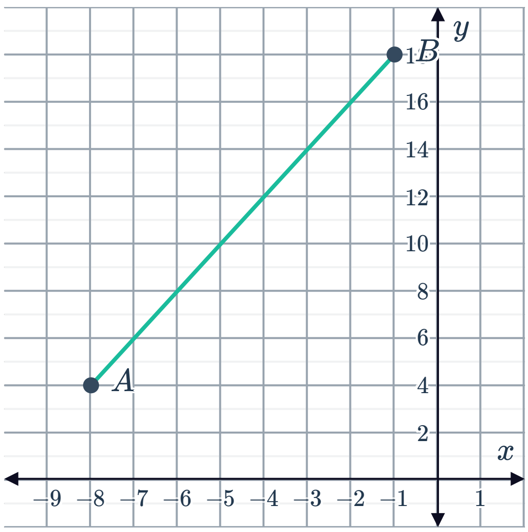
a $A\,(2, 4)$ and $B\,(8, 6)$



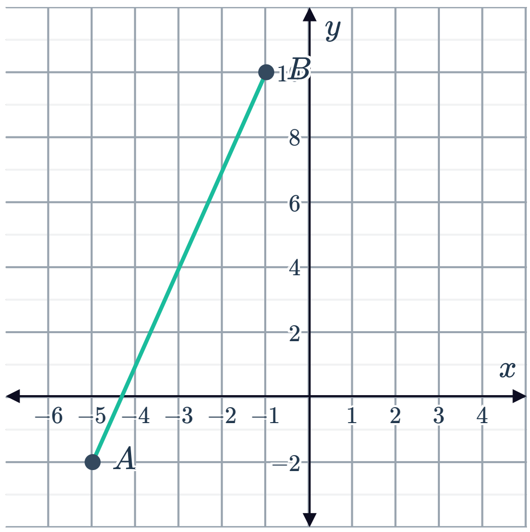
b $A\,(-3, 4)$ and $B\,(3, 16)$



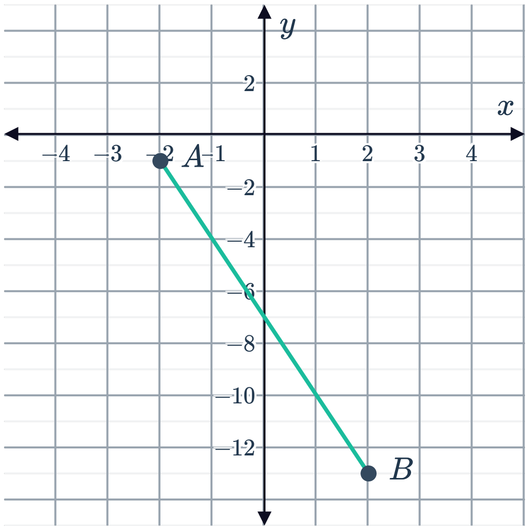
c $A\,(-8, 4)$ and $B\,(-1, 18)$



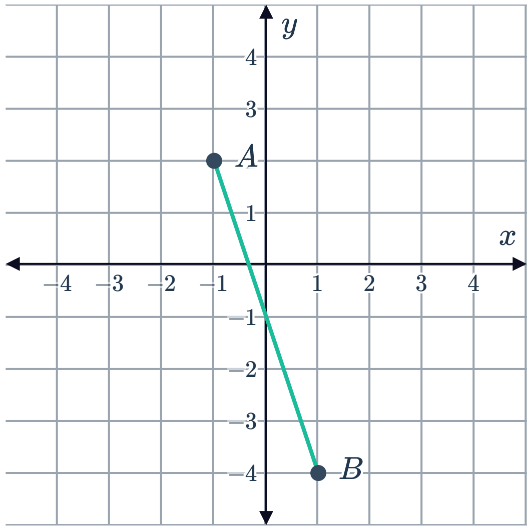
d $A\,(-5, -2)$ and $B\,(-1, 10)$



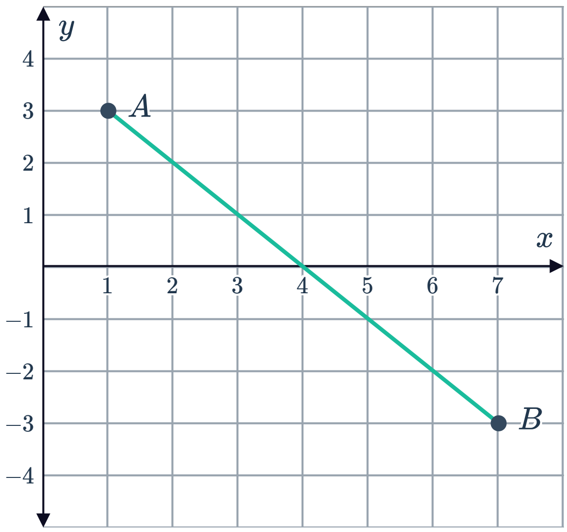
e $A\,(-2,-1)$ and $B\,(2,-13)$



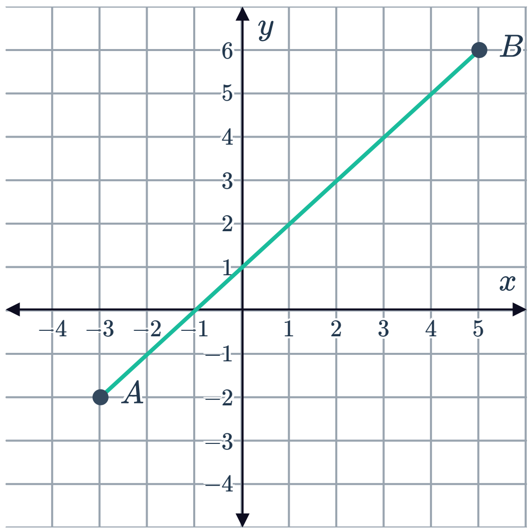
f $A\,(-1,2)$ and $B\,(1,-4)$



g $A\,(1,3)$ and $B\,(7,-3)$



h $A\,(-3,-2)$ and $B\,(5,6)$

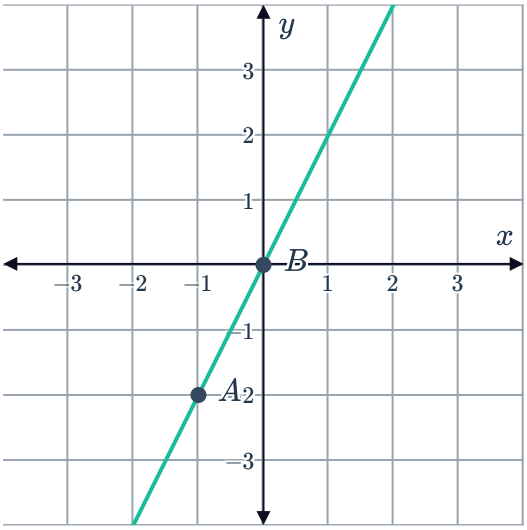


3 Explain why a vertical line has an undefined gradient.

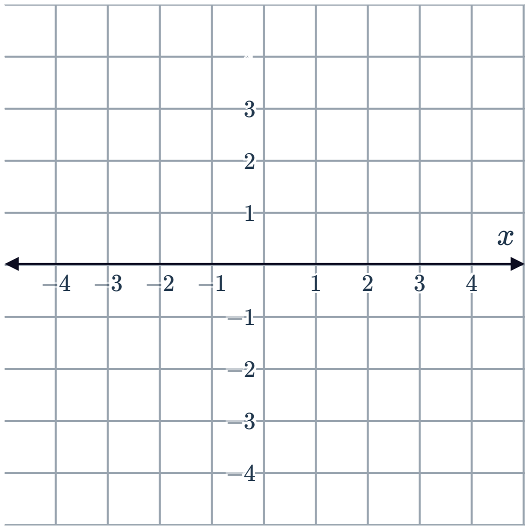
4 State the gradient of any line parallel to the x-axis.

5 Find the gradient of the following lines passing through points A and B:

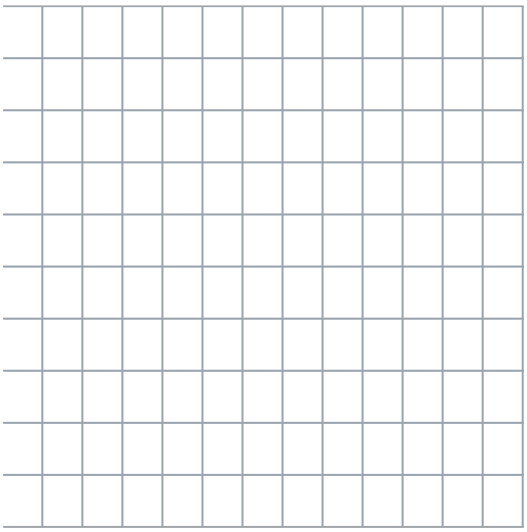
a



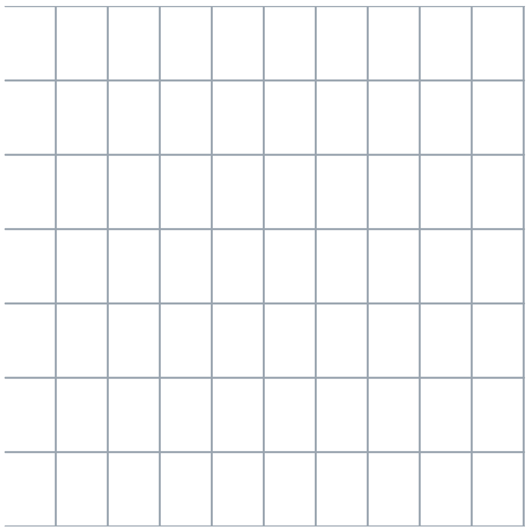
b



c



d



Gradient formula

6 Find the gradient of the interval joining $A\,(-9,4)$ and $B\,(-3,-5)$.

7 If we have two points and the slope formula $m=\dfrac{y_2-y_1}{x_2-x_1}$, does it matter which point is (x_1,y_1) and which point is (x_2,y_2)?